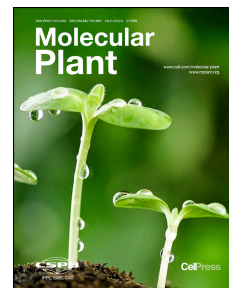


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AI-driven protein engineering: A new paradigm for plant trait design

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Short Summary

This review examines how artificial intelligence is transforming protein engineering for plant trait design by integrating structure prediction, generative sequence modeling, and function optimization to expand the accessible design space for proteins with enhanced agronomic performance. It synthesizes key application scenarios, current translational bottlenecks, and a staged roadmap toward deployable AI-driven breeding strategies.

22 **Abstract**

23 Protein engineering modifies protein molecules to achieve specific biological or
24 technological functions. Protein design forms the core methodology, and in recent
25 years, artificial intelligence (AI)-driven approaches have enabled more precise trait
26 design in plants. This review highlights the convergence of protein structure
27 prediction, generative sequence modeling, and function optimization to create
28 synthetic proteins with improved specificity, stability, and activity in plant systems.
29 We trace the development of protein design, from rational design to semi-rational
30 strategies and AI-driven platforms that integrate structure prediction, sequence
31 generation, and *de novo* design. We discuss eight application areas relevant to plant
32 physiology and breeding: enhanced disease resistance via engineered immune
33 receptors; insect resistance through optimized insecticidal proteins; abiotic stress
34 tolerance through metabolic enzyme stabilization; improved nutrient use via
35 transporter redesign; variant mining for trait fine-tuning; genome editing system
36 optimization; environmental sensing with synthetic biosensors; and programmable
37 regulatory circuits for plant factories (including controlled environment agriculture).
38 Across these areas, we summarize design principles, advances, and translational
39 considerations, emphasizing how AI expands sequence space and improves candidate
40 prioritization. We also address current bottlenecks, including domain shift, generative
41 model reliability gaps, limited portability, the genotype-to-phenotype gap, and design-
42 to-validation workflow constraints. Finally, we propose a staged roadmap for AI-
43 driven plant trait design, outlining milestones and requirements for translation into
44 breeding.

Introduction

Proteins carry out most cellular functions and translate genetic information into the structural, catalytic, regulatory, and signaling processes that govern plant growth, development, and adaptation. In crops, key proteins, including immune receptors, metabolic enzymes, transcriptional regulators, and membrane transporters, directly affect yield potential, stress resilience, nutrient use efficiency, and product quality (Sedeek et al., 2019). For decades, plant breeding has leveraged natural variation in these proteins through selection and hybridization. However, many potentially valuable sequence and structural variants remain inaccessible to conventional breeding, which relies largely on naturally occurring diversity and empirical screening (Phillips, 2010).

To avoid ambiguity and clarify the framework of this review, we define key terms first. Protein engineering modifies protein molecules to achieve specific biological or technological functions (Lutz and Iamurri, 2017). It relies on the sequence to structure to function relationship and integrates genetic, chemical, and computational approaches to tailor protein properties toward defined goals (Lutz and Iamurri, 2017). Within this framework, protein design provides a core methodology (Bornscheuer and Pohl, 2001). Protein design draws on models of folding and molecular interactions described by free energy landscapes and aims to infer or optimize amino acid sequences that produce desired structural or functional states (DeGrado et al., 1989). In this review, we define AI-driven protein engineering approaches as those in which AI autonomously generates or optimizes sequences and structures based on large-scale data and models. In contrast, AI-supported refers to the role of AI in assisting human decision-making by supporting tasks such as structure modeling, candidate prioritization, and mutational effect prediction, without fully replacing human input. These terms are used throughout the manuscript to distinguish between fully autonomous workflows (AI-driven) and supportive AI applications (AI-supported), which aid but do not replace human expertise in protein design and engineering.

In practice, rational design refers to strategies that use known or reliably modeled three-dimensional structures and reaction mechanisms. Rational design applies physicochemical principles to introduce a small number of targeted mutations or to build new sequences in a directed manner (Hellinga, 1997). Rational design is interpretable when molecular determinants are well defined, but it often explores a restricted design space because it depends on strong prior assumptions. Semi-rational design combines elements of rational design and directed evolution (Chica et al., 2005). Key residues are selected based on prior knowledge, then subjected to combinatorial mutagenesis followed by screening or selection. Semi-rational design can explore sequence space efficiently while maintaining a bias toward functional relevance (Chica et al., 2005; Lutz, 2010). In practice, rational and semi-rational strategies are often combined, forming a continuum that can be adapted to different biological contexts and application goals.

Building on these foundations, advances in artificial intelligence (AI) and protein science are enabling a shift from discovery to deliberate construction of functional proteins (Yang et al., 2019). Deep learning-based structure prediction, generative sequence modeling, and computational function optimization expand exploration of sequence space beyond the constraints of natural evolution and support targeted engineering of molecular properties (Jumper et al., 2021). These developments make it increasingly feasible to reprogram plant proteins toward breeding goals, including improved enzymatic efficiency, altered binding specificity, enhanced stability under stress, and tuned regulatory dynamics (Voytas and Gao, 2014). Machine learning approaches also integrate multi-omics data to accelerate gene discovery and trait optimization, which can complement protein design workflows in breeding programs (Cheng and Wang, 2024). Together, these advances can shorten development cycles and reduce reliance on trial and error. AI-driven protein design is not simply automation of rational design; it adds statistical learning as a central modeling principle.

Protein design methods have progressed toward greater efficiency, scalability, and predictive performance as computational tools and data resources have improved. Figure 1 summarizes this trajectory. Rational design, semi-rational design, AI-driven approaches, and de novo design do not replace one another. These approaches coexist in practice and together form a methodological framework for modern protein engineering (Lutz and Iamurri, 2017).

In this review, we examine how AI-driven protein engineering can contribute to plant breeding. We first describe the evolution of protein design approaches, from rational and semi-rational strategies grounded in structural biology and directed evolution to data-driven approaches supported by AI. We then discuss eight application scenarios relevant to plant physiology and breeding: disease and pest resistance, abiotic stress resilience, nutrient and resource use efficiency, variant mining for trait fine-tuning, optimization of genome editing components, environmental sensing, and adaptation to controlled environment agriculture. We also discuss bottlenecks that currently limit broader application in plant systems and outline directions to address them. Finally, we present a staged roadmap for plant trait design using AI-driven protein engineering, with realistic milestones that connect near-term decision support to longer-term breeding platforms grounded in plant-specific constraints and governance.

The evolution of protein design technologies

Protein design aims to create or optimize protein sequences with defined structural or functional properties (Listov et al., 2024). In plants, this can involve modifying endogenous proteins or introducing synthetic components to reprogram signaling, metabolism, regulation, or interaction networks (Lyu et al., 2021). Over the past two decades, protein design has progressed from low-throughput, trial-and-error exploration to a more systematic discipline supported by structural biology, computational modeling, and data-driven methods.

Early work relied mainly on rational design, where experimentally determined structures and biochemical knowledge guided targeted interventions such as site-directed mutagenesis and domain engineering (Hellings, 1997). These approaches provided important advances but were constrained by limited structural coverage, especially for many plant proteins, and by labor-intensive validation. Rational design also required substantial expertise and iterative testing, which makes it well suited to well-characterized targets but less efficient for new or complex applications in plant biology (Listov *et al.*, 2024).

To address these limitations, computational modeling and higher-throughput experiments supported the rise of semi-rational design. Semi-rational approaches combine rational guidance with directed evolution and mutational scanning to explore protein fitness landscapes more systematically. Structure-based platforms such as Rosetta support backbone sampling and sequence optimization using known or predicted structures, enabling refinement of existing folds while incorporating biophysical and evolutionary constraints (Abbass and Nebel, 2020). In plant systems, semi-rational strategies remain largely at the proof-of-concept stage, but they have improved native enzymes, receptors, and immune sensors. Reported outcomes include higher catalytic efficiency, increased thermostability, and altered ligand specificity in hormone receptors and nucleotide-binding leucine-rich repeat proteins (Cadiou *et al.*, 2023; Lozano-Juste *et al.*, 2023; Zdrzałek *et al.*, 2023).

More recently, AI tools have increased both the precision and the scalability of protein design. Structure predictors such as AlphaFold and RoseTTAFold have expanded structural coverage of plant proteomes (Ceasar and Ebeed, 2024; Lin *et al.*, 2025). This broader structural landscape supports downstream design and functional optimization and can help prioritize variants with higher designability and evolvability (Freschlin *et al.*, 2022). Sequence-based tools, including ProteinMPNN and ESM, support generative sampling of variants, often coupled with directed evolution to test and refine designs (Dauparas *et al.*, 2022; Lin *et al.*, 2023; Madani *et*

al., 2023; Mardikoraem et al., 2023). Protein language models (PLMs) extend this framework by using large-scale sequence data to support data-driven modification and prediction of mutational effects (Ferruz and Höcker, 2022). By modeling fitness landscapes, PLMs can estimate how amino acid changes affect stability, activity, or interactions. These predictions can guide targeted mutations to improve enzyme kinetics or receptor affinity in plant systems (Meier et al., 2021). PLMs also support redesign of gene editors, including Cas proteins, to improve specificity, reduce off-target activity, or enhance compatibility with plant cells (He et al., 2024). In addition, PLMs can support trait fine-tuning through iterative refinement of sequences to achieve modest shifts in response behavior, such as adjusting stress-response thresholds or metabolic flux while maintaining core function (Yoosefzadeh-Najafabadi, 2025). Overall, PLM-based workflows link prediction with experimental validation, accelerating design cycles and supporting applications such as adapting microbial effectors or genome-editing nucleases for expression in plants (Motmaen et al., 2023; Yang *et al.*, 2019).

Looking ahead, the field is moving toward *de novo* design, where models generate novel protein architectures without relying on natural templates, with the potential to expand the range of achievable functions (Supplemental Figure 1). Diffusion-based approaches such as RFdiffusion and Chroma support end-to-end design of backbones and side chains conditioned on structural or functional goals (Singh, 2024; Yim et al., 2024). Current applications focus mainly on relatively simple scaffolds or binders, but improvements in model performance and integration of multiscale constraints may extend these approaches to more complex enzymes, oligomers, and synthetic modules (Fox et al., 2025). These advances are expected to integrate with iterative design-build-test-learn pipelines, enabling programmable protein-based solutions for crop improvement (Gurdo et al., 2023; Xiong et al., 2021). As AI methods better integrate evolutionary information with biophysical constraints, *de novo* design may enable traits that are difficult to reach with earlier approaches (Kortemme, 2024).

Application scenarios of protein engineering in plant trait design

Protein engineering is increasingly used for crop improvement because it provides a systematic way to manipulate biological function beyond the limits of natural variation (Li et al., 2025a). In plant systems, much of the successful work has relied on structure-guided design and mutagenesis, with advances in disease resistance, insect resistance, metabolic optimization, and molecular sensing (Beltrán et al., 2022; Chougule et al., 2013; De la Concepcion et al., 2019; Zhao et al., 2024). Most of these studies fall within rational or semi-rational paradigms, and deep integration of AI into protein engineering workflows remains at an early stage in plant biology, with relatively few examples where AI directly generates sequences or structures. Even so, recent progress in AI-driven structure prediction, generative modeling, and sequence function learning has created opportunities to extend validated strategies into larger design spaces (Liu et al., 2025). Here, we review proof-of-principle studies based on rational and semi-rational protein design across eight representative application scenarios in plant breeding (Figure 2). We then discuss how AI-based approaches could support these efforts through design principles and workflow organization.

Enhancing disease resistance

Plant immunity relies on receptor-mediated recognition, signal transduction, and effector-triggered responses that protect crops from pathogens. Traditional breeding for disease resistance has relied largely on naturally occurring immune receptors, especially nucleotide-binding leucine-rich repeat proteins, which activate defense after detecting pathogen effectors (Barragan and Weigel, 2021). However, rapid effector evolution, narrow recognition ranges of natural receptors, and limited durability of single-gene resistance under field conditions remain major challenges. Protein engineering can reprogram immune components to broaden recognition, increase durability, and tune activation behavior (Marchal et al., 2022). Many studies show that rational design guided by structural and mechanistic hypotheses can

improve resistance by modifying nucleotide-binding leucine-rich repeat domains or by building chimeric receptors, providing a foundation for more systematic immune protein engineering.

Expanding effector recognition is a central goal. In rice (*Oryza sativa*), structure-guided engineering of the Pikp receptor, supported by crystal structures and AlphaFold-Multimer models, broadened recognition of multiple AVR-Pik effector variants (De la Concepcion *et al.*, 2019; Maidment *et al.*, 2023). In tomato (*Solanum lycopersicum*), targeted mutagenesis of the LRR domain in Tm-2² restored binding to the movement protein of tomato brown rugose fruit virus and conferred stable resistance to this virus and related tobamoviruses using protein engineering techniques (Wang *et al.*, 2025a). AI-supported structural modeling and mutational effect prediction were used to assist in guiding the design of these mutations, complementing classical mutagenesis techniques. Together, these studies show that directed remodeling of receptor-effector interfaces can provide a predictable route to engineered disease resistance.

Beyond recognition, protein design has improved regulation of immune signaling. In tobacco and soybean, insertion of conserved viral protease cleavage sites into autoactive nucleotide-binding leucine-rich repeat proteins enabled pathogen-dependent activation and conferred broad resistance to multiple potyviruses (Wang *et al.*, 2025b). Structure-guided mutations in helper nucleotide-binding leucine-rich repeat proteins can also relieve suppression by pathogen effectors within signaling modules, restoring immune activation and resistance mediated by several sensor receptors (Contreras *et al.*, 2023). These examples indicate that tuning conformational state and activation thresholds can increase defense output without changing recognition specificity.

Protein design can also act on non-receptor components to strengthen basal defenses or remove immune suppressors. In common bean, structure-guided optimization of polygalacturonase-inhibiting protein 2 increased inhibition of fungal

polygalacturonases, which promoted defense signaling and provides a blueprint for receptor-independent defense modules (Xiao et al., 2024). A targeted condensate degradation system, created by fusing high-affinity effector-binding domains to ubiquitin E3 ligase modules, enabled removal of immune suppressors in rice. Under pathogen-inducible promoters, this strategy increased resistance to rice blast disease while preserving normal growth and has been validated under field conditions (Luo et al., 2025; Yan et al., 2024).

Together, these studies show that protein engineering can improve plant disease resistance through interventions at key immune nodes that act during pathogen challenge. These interventions include reshaping receptor-effector interfaces, tuning conformational state and activation thresholds within signaling modules, and strengthening basal defenses or removing pathogen effectors or immune suppressors. Effective engineering often requires balancing immune efficacy with regulatory stability and evolutionary robustness, rather than maximizing a single molecular property. Most current examples still rely on rational and semi-rational strategies, with data-driven approaches used mainly for structure modeling, interface evaluation, and variant prioritization. As larger mutation-to-phenotype datasets become available in plants, protein design for disease resistance should become more efficient and scalable.

Reinforcing pest resistance

Protein-based insecticidal agents remain central to crop pest control. Among these agents, *Bacillus thuringiensis* (Bt)-derived Cry proteins form the cornerstone of many transgenic insect-resistant crops. Cry proteins bind specific receptors, including alkaline phosphatase and ABC transporters, on the brush border membrane of insect midgut cells. Binding triggers pore formation, which causes osmotic imbalance, cell lysis, and ultimately insect death. However, long-term field deployment has selected for resistance through mechanisms such as receptor gene mutations, transcriptional downregulation, and altered gut protease activity, which together reduce the efficacy

of Bt toxins (Jurat-Fuentes et al., 2021). Protein engineering can help mitigate resistance and broaden insecticidal spectra.

Structure-guided rational design has improved Cry protein performance. Targeted mutations in key regions, including domain I alpha helices, domain II loops (L1 to L3), and domain III β -sheets, can increase activity, expand host range, and improve stability under stresses such as ultraviolet exposure (Vilchez, 2020). One example is GEAb-dVL, an engineered antibody scaffold derived from the Cry1Ab template. Structural modeling and docking were used to design binding to midgut membrane proteins of *Plutella xylostella*, including the V-ATPase A subunit and ABC transporters. Unlike classical Cry proteins, GEAb-dVL induces mitochondrial-mediated apoptosis through a non-canonical pathway, which points to antibody-like scaffolds as programmable platforms for toxin design (Xie et al., 2021).

Protein engineering has also extended beyond Cry proteins to other Bt-derived toxins, including Vip, Cyt, and Sip proteins, which broadens the diversity of insecticidal molecules and their modes of action (ChongXin et al., 2024). Engineering of natural peptides, including peptides from spider venom, can improve stability and yield mimetics such as kinin analogs with strong insecticidal activity and low toxicity, supporting reduced-risk pesticide development (Zhang et al., 2023b). More recently, deep learning-based sequence analysis has been used to prioritize candidate peptides from metagenomic datasets. Coupled with in vitro validation of insecticidal or antimicrobial activity, these approaches can increase the efficiency of functional peptide discovery (Ma et al., 2022; Santos-Júnior et al., 2024).

Looking ahead, modular and programmable strategies may expand the design space for plant insect resistance. One route is to assemble synthetic proteins that combine domains for receptor binding, membrane disruption, and intracellular interference into multicomponent molecules with broad activity and layered mechanisms.

Environment-responsive regulatory elements, including wound-inducible promoters or herbivore-induced volatile sensors, could enable spatiotemporally controlled

expression of insecticidal proteins in planta, improving precision and biosafety (Turlings and Erb, 2018). Data-driven tools for structure prediction and functional motif design also support development of toxin classes with new modes of action (Kong et al., 2024). Another emerging strategy is plant-expressed biological proteolysis targeting chimeras (bioPROTACs), which fuse a binding module with a ubiquitin ligase recruiting domain to drive degradation of essential insect proteins in the gut or nervous system (Békés et al., 2022). Such programmable molecules could provide selective and potent interventions with reduced off-target effects.

Overall, protein engineering for insect resistance has focused on remodeling toxin receptor interactions, diversifying modes of action, and refining regulatory control. Durable insect resistance is likely to depend on coordinated tuning of toxic effects, target range, and environmental robustness. Most current work remains grounded in rational or semi-rational frameworks, while data-driven methods are used mainly for structure function analysis and candidate screening to reduce experimental search space. As functional datasets grow and mechanisms underlying durability and ecological adaptation become clearer, insect resistance engineering should become more predictable and easier to translate.

Boosting abiotic stress resilience

Global climate change is increasing the frequency and complexity of abiotic stresses, including drought, temperature extremes, salinity, and osmotic stress. These stresses reduce yield and quality by disrupting key physiological processes such as stomatal regulation, osmotic adjustment, ion homeostasis, and metabolism. Stress perception and response rely on complex signaling and transcriptional networks (Zhang et al., 2022). Because available natural variation is often limited, breeding programs aim to improve stress tolerance while preserving growth and resource use efficiency. Protein engineering offers a way to intervene in key stress-response components and to tune the intensity and spatial and temporal dynamics of stress responses.

Protein engineering has been used to modify proteins within stress-response pathways to improve performance under adverse conditions. Structure-guided engineering of the abscisic acid receptor PYR1 enabled recognition of exogenous ligands and activation of drought-response signaling, which improved drought tolerance under chemically induced conditions (Park et al., 2015). This work shows that receptor reprogramming can be used to modulate stress pathways and provides a basis for developing tunable environmental response mechanisms. In maize, fusion of a heat-stable cytosolic isozyme with a plastid targeting peptide produced an engineered PGD variant that retained activity at elevated temperature and sustained starch biosynthesis during kernel development (Ribeiro et al., 2020). This example illustrates how stability engineering of metabolic enzymes can mitigate temperature stress and suggests an approach that may translate across species and metabolic contexts. Engineering of components in the salt overly sensitive pathway has also been reported to improve sodium and potassium homeostasis under salinity, which can reduce growth penalties caused by salt stress (Zhang et al., 2023a).

Protein engineering has also been applied to oxidative and low temperature stress. Directed evolution of NADPH-dependent reductases and engineered oxidase variants can improve redox balance and reactive oxygen species scavenging, reducing cellular damage under oxidative stress (Maxel et al., 2020). For cold stress, most work has focused on genetic changes in regulatory networks and transcription factors, but several results inform future protein engineering. For example, increased activity or stability of CBF or DREB transcription factors can improve cold tolerance in crops (Ding et al., 2024b). In rice, fine-tuned regulation of *OsDREB1C* increased chilling tolerance and, under specific conditions, maintained photosynthetic performance and yield traits (Wei et al., 2022), which suggests that such regulators are promising targets for future engineering.

As computational tools and synthetic biology continue to improve, protein engineering is also moving toward more system-level control of stress responses. By

combining environmental sensing modules, programmable regulatory elements, and synthetic control circuits, researchers have begun to explore designs that coordinate responses to multiple abiotic cues. Most examples remain proof of concept, but available studies support the feasibility of reshaping stress-response networks at the whole plant scale (Kumar et al., 2023).

Overall, protein engineering for abiotic stress tolerance has focused on tuning stress-response networks, preserving function of key proteins under adverse conditions, and improving cellular homeostasis. These strategies usually aim for coordinated regulation across growth, metabolism, and adaptation rather than maximizing a single molecular property. Most current studies rely on rational or semi-rational design, with data-driven methods used mainly to identify high-leverage nodes within stress networks and to prioritize feasible engineering strategies. For breeding, evaluation across environments and genetic backgrounds remains essential to assess stability, metabolic costs, and trade-offs, and to improve field-level predictability and translation.

Improving efficiency and nutrition

Constraints on agricultural resources and increasing environmental pressure have made improved uptake and use of carbon, nitrogen, and mineral nutrients a major goal in molecular breeding. Approaches that rely mainly on natural variation or transcriptional regulation often do not resolve systemic bottlenecks such as coordination across enzymes, coupled fluxes, and environmental heterogeneity. Protein engineering can intervene at key nodes in metabolic networks and optimize resource-use pathways, shifting effort from improving single enzymes to redesigning pathway performance for higher efficiency and nutritional quality. This section summarizes advances in protein engineering, with emphasis on rational and semi-rational strategies that improve crop performance and nutritional value.

Improving carbon use is central to higher photosynthetic productivity, and Rubisco is a key target because of low catalytic efficiency and high oxygen affinity (Prywes et

al., 2023). Rational design and directed evolution have improved catalytic properties of Rubisco large and small subunits (Gionfriddo et al., 2024). Protein engineering has also supported introduction of carbon concentrating mechanisms from algae and cyanobacteria into plant chloroplasts, which can increase carbon dioxide availability (Fei et al., 2022; Flecken et al., 2020). Synthetic biology has advanced artificial carbon fixation cycles, including the CETCH and MOG cycles, which aim to exceed constraints of native photosynthesis (Ye et al., 2024). Challenges remain in expression control and subcellular targeting, but together these studies show the potential of protein engineering to build cross-species carboxylation systems and associated regulatory components.

Protein engineering is also being used to improve nitrogen and phosphorus efficiency. For nitrogen, a long-term goal is to confer nitrogen fixation capacity to crops. Structural optimization of nitrogenase components, together with controlled localization and coordination with host metabolism, can improve stability and catalytic performance of nitrogen fixing rhizosphere microorganisms and strengthen plant microbe functional interactions (Haskett et al., 2021). Although transfer of symbiotic nitrogen fixation to non-legumes remains challenging (Pankievicz et al., 2019), rational design of domain architecture, transport signals, and cofactor compatibility could support steps toward nitrogenase function in plant mitochondria or plastids. For phosphorus, rational modification of phosphate transport proteins has been used to improve uptake. Site-directed mutagenesis of Arabidopsis PHT1;1 increased phosphate transport activity, providing direct evidence that transporter function can be optimized through protein engineering (Fontenot et al., 2015).

Beyond resource-use efficiency, protein engineering combined with synthetic metabolic engineering is improving nutritional quality in staple crops. Targeted accumulation of health-promoting pigments, including carotenoids, astaxanthin, and anthocyanins, together with reduction of anti-nutritional factors such as phytate, can improve nutritional profiles (Zhang et al., 2025). Rational redesign and directed

evolution of key enzymes have also increased vitamin biosynthesis, including folate, and enabled introduction of pathways for unsaturated fatty acids that are limited or absent in many plants (Zhu et al., 2020). Examples include Golden Rice 2, which introduced *phytoene synthase* to increase beta carotene (Paine et al., 2005); transfer of bacterial *crtB* and *crtI* into wheat grain to increase carotenoids (Wang et al., 2014); and expression of *GTP cyclohydrolase I* and *aminodeoxychorismate synthase* to raise folate by about 100-fold in rice (Storozhenko et al., 2007).

In summary, rational and semi-rational protein engineering can improve resource-use efficiency and nutritional value by enhancing metabolic enzymes, optimizing nutrient uptake systems, and reconfiguring regulatory control within key pathways. Looking ahead, integration of modular metabolic parts with machine learning informed systems models and quantitative phenotyping could support programmable control of metabolism and nutrition. Such approaches may enable more efficient and stable trait outcomes under defined resource constraints.

Variant mining and trait fine-tuning

Identifying and deploying functional genetic variants is essential for improving complex agronomic traits. Traditional molecular breeding relies on natural or mutagenized populations, using quantitative trait locus mapping and genome-wide association studies to identify candidate genes, followed by functional validation and allele introgression. These approaches offer limited tunability and can be inefficient for polygenic traits with context-dependent outcomes. Protein engineering provides an alternative by moving from random variant screening to structure-informed, function-guided fine-tuning. Here, we summarize recent progress in protein engineering and data-driven prediction tools for variant mining and precise trait modulation.

Classical mutagenesis, using physical treatments such as gamma rays or chemical treatments such as ethyl methanesulfonate, introduces genome-wide mutations but often yields many low-relevance variants and substantial background noise, which

complicates identification of useful alleles (Shirasawa et al., 2016). Even when functional mutants are recovered, phenotypes are often strong or reflect complete loss of function. Such alleles can be useful for gene discovery but may lack the precision needed for breeding. For example, mutations in maize *brachytic2*, which encodes a P-glycoprotein auxin transporter, cause dwarfism but also reduce ear height and internode length, which can compromise yield (Multani et al., 2003). These outcomes illustrate the need for milder perturbations that optimize traits without disrupting plant architecture.

Mutation effect prediction offers a computational route to higher-throughput variant mining by prioritizing amino acid substitutions that are more likely to affect conformation, substrate binding, or stability. Methods such as DeepDDG (Cao et al., 2019), PPVED (Gou et al., 2022), and SaProt (Su et al., 2023), together with structure prediction models such as AlphaFold (Jumper *et al.*, 2021) and RoseTTAFold (Baek et al., 2021), can support prioritization of substitutions expected to alter structure or function. PPVED has been validated on proteins from Arabidopsis and maize for predicting functional effects of stress-related single amino acid substitutions (Gou *et al.*, 2022). In addition, combining free energy change calculations with molecular docking and molecular dynamics simulations can connect mutations to stability and binding changes at a mechanistic level and guide selection for downstream testing (Buel and Walters, 2022; Sora et al., 2023). This workflow can enrich for mild, attenuating variants that tune activity, binding specificity, or signaling sensitivity while preserving core function, which is often desirable for trait optimization.

For targeted construction of allelic variants, protein design can be coupled with base editing to introduce single nucleotide substitutions in a controlled way. Compared with promoter editing or knockout strategies, structurally guided selection of candidate substitutions can help identify residues where modest changes adjust enzyme activity, transcription factor DNA binding specificity, or signaling sensitivity while maintaining folding and overall function. One example is AiCE, which

integrates structural and evolutionary constraints with inverse folding models. AiCE has been used to optimize deaminases, nuclear localization signals, nucleases, and reverse transcriptases, supporting development of improved base editors for applications in agriculture and medicine (Fei et al., 2025).

Overall, work in this area can be grouped into three linked components: variant discovery, effect prediction, and targeted construction. Population resources and mutagenesis support discovery of candidate alleles. Sequence and structure models support prioritization, and genome editing tools, including base editing, enable controlled introduction when needed. AI-based methods can accelerate effect prediction and screening by learning complex sequence to phenotype relationships, which narrows the scope of experimental validation. With larger and higher-quality mutation, structure, and phenotype datasets and validation across diverse genetic backgrounds, predictive reliability should improve and translation of controllable allelic variants into breeding should accelerate.

Optimizing genome editing components

Building on site selection and variant prioritization, genome editing systems determine whether candidate designs can be installed in plant genomes with high fidelity, predictable outcomes, and stable inheritance. Over the past decade, engineering of CRISPR-associated proteins has improved specificity, efficiency, and programmability through changes to catalytic cores, DNA recognition interfaces, and RNA-associated modules. These advances have expanded the range of editable loci and enabled more controlled modification of agronomic traits (Li et al., 2025b; Ruffolo et al., 2025).

Structure-guided engineering of Cas nucleases, including *Streptococcus pyogenes* Cas9 and *Lachnospiraceae bacterium* Cas12a, has improved cleavage fidelity and expanded targeting. Mutations that reduce nonspecific DNA contacts have produced high-fidelity variants, including SpCas9-HF1 (Kleinstiver et al., 2016) and eSpCas9

(Slaymaker et al., 2016), which lower off-target activity while retaining practical on-target efficiency. In plants, these variants have been used for multiplex editing in rice, with reduced off-target effects and usable editing efficiencies (Kulcsár et al., 2017; Zhang et al., 2017). Engineering of protospacer adjacent motif recognition has also relaxed PAM constraints and expanded targetability. For example, engineered Cas12a variants such as LbCas12a-RVR and enAsCas12a broaden PAM compatibility and enable editing at loci that were previously inaccessible (Toth et al., 2020).

Base editors enable single-nucleotide conversion by fusing a deaminase to a Cas9 nickase or catalytically inactive Cas protein, which avoids double-strand breaks. Cytosine base editors (Komor et al., 2016) and adenine base editors (ABEs) (Gaudelli et al., 2017) established the core architectures. Engineering of deaminase domains, including APOBEC1 and TadA variants, together with optimization of Cas9 nickase scaffolds, has improved precision, reduced by-products, and expanded sequence compatibility. Recent developments include Cas-independent deaminase DNA-binding fusions and compact editors designed for delivery constraints (Porto and Komor, 2023). In rice, ABE8e enabled efficient A to G editing, supporting precise nucleotide substitutions in genes linked to herbicide resistance and yield traits without introducing double-strand breaks (Li et al., 2024).

Prime editing combines a Cas9 nickase (H840A) with a reverse transcriptase and uses a prime editing guide RNA (pegRNA) that encodes both the target site and the desired sequence change. Prime editing supports base substitutions and small insertions or deletions and is widely used as a versatile framework for precise genome modification (Anzalone et al., 2019). In plants, prime editing has been implemented in protoplasts and regenerated plants of crops such as rice and wheat, enabling point mutations and short indels (Lin et al., 2020), and has achieved stable, precise edits in rice cells (Tang et al., 2020). In maize, improved pegRNA expression and dual pegRNA vector designs enabled W542L and S621I substitutions in *ZmALS1* and *ZmALS2*, generating herbicide-resistant lines (Jiang et al., 2020). Higher-efficiency prime editing variants,

including the PE6c variant, have further improved editing efficiency across multiple targets in monocot crops (Cao et al., 2024). Strategies such as DualPE also extend plant prime editing to support deletion, replacement, and inversion of larger DNA fragments, which expands the toolkit for complex trait engineering and chromosome-scale modification (Zhao et al., 2025). However, broader validation across target loci and genetic backgrounds is still needed to assess robustness in breeding contexts.

Beyond sequence changes, modular genome editing systems enable programmable regulation of gene expression. PAM-flexible Cas variants combined with designed guide RNAs support epigenome editing and transcriptional reprogramming without altering DNA sequence. In *Arabidopsis*, the SunTag system reactivated the silenced gene *FWA* and provided strong transcriptional activation for studies of epigenetic regulation and development (Papikian et al., 2019). Crop engineering often also requires spatiotemporal control to reduce growth penalties and limit unintended phenotypes. Chemically inducible and condition-responsive strategies are therefore relevant for translation. For example, the ER-Tag system provides chemically inducible control and improved regeneration in several species, including alfalfa, woodland strawberry, and sheepgrass, by activating endogenous morphogenic genes (Zhang et al., 2024).

In summary, engineering of genome editing components has shifted from enabling editability to improving precision, controllability, and translation. Nuclease engineering improves fidelity and expands targetable sequence space, base editing increases predictability of single-nucleotide changes while reducing by-products, and prime editing expands the capacity for precise sequence writing and more complex modifications. At present, AI is most useful for guide and pegRNA design, off-target assessment, prediction of editing outcome distributions, and screening of construct combinations to reduce experimental iteration. A practical path forward is to build on engineered editing platforms while adding data-driven parameter optimization and

standardized phenotypic evaluation, which can speed closed-loop iteration from molecular edits to trait-level gains.

Programming environmental sensing systems

Plants continuously monitor internal and external cues, including temperature, light, water status, nutrient availability, and biotic and abiotic stress. Approaches that rely mainly on downstream transcriptional or phenotypic outputs often lack the temporal resolution and subcellular specificity needed to dissect signal transduction. Protein engineering can create synthetic sensors that decode environmental signals with high spatiotemporal precision and support dynamic, programmable responses in plants. Here, we summarize recent advances in engineering environmental sensing systems and discuss relevance for plant bioengineering and trait regulation. In parallel, high-throughput phenotyping platforms that combine imaging with machine learning have been used for real-time analysis of physiological responses to salt and drought stress. These platforms translate complex environmental inputs into computable phenotypic features and provide early examples of workflows that incorporate environmental sensing at scale (Feng et al., 2020).

Protein-based biosensors enable real-time tracking of environmental and molecular cues in plants. AuxSen is a synthetic auxin sensor that combines an engineered TrpR domain with a fluorescence resonance energy transfer readout to monitor indole-3-acetic acid dynamics during root gravitropism with reversible, high-resolution responses (Herud-Sikimić et al., 2021). Although demonstrated mainly in laboratory models, robust performance in a complex organism supports broader physiological and agronomic applications. PAleon uses conformational selection to report subcellular distributions of phosphatidic acid and revealed salt stress induced relocalization between the plasma membrane and endoplasmic reticulum (Li et al., 2019). Together, these sensors show how structure-guided design of modular components can achieve specific molecular recognition and reliable signal transduction.

Beyond sensing, protein engineering can reprogram natural receptors and create artificial binders to expand the range of detectable inputs. Structure-guided modification of the abscisic acid receptor PYR1 enabled recognition of non-native small molecules, demonstrating that endogenous plant receptors can serve as adaptable sensing scaffolds (Leonard et al., 2024). Repeat-based scaffolds and pseudo-cyclic backbones have also been developed as small-molecule binders with tunable affinity and selectivity, which supports higher-throughput ligand recognition and sensor development (An et al., 2024).

A further step is integration of sensing and actuation within a single module to enable one-step signal transduction. Glasgow et al. engineered an allosteric sensor for farnesyl pyrophosphate using a split reporter, coupling ligand binding at a designed dimer interface to output amplification with near-atomic control (Glasgow et al., 2019). In membrane-localized systems, synthetic four-helix bundles such as tmFAP have been tuned for selective small-molecule binding through hydrophobic surface engineering. Targeting to membrane microdomains with glycosylphosphatidylinositol motifs can further improve spatial resolution (Zhu et al., 2025). Together, these advances suggest that tight coupling of recognition and output within a single module provides a tractable route toward closed-loop response units in plant systems.

Overall, environmental sensing engineering extends plant trait improvement from static genetic modification toward dynamic regulation. Engineered sensors increase spatiotemporal resolution for monitoring stress and developmental signals, while receptor reprogramming and integrated sensor-effector modules expand controllable inputs and outputs. For crop improvement, these systems need evaluation across genetic backgrounds to assess stability, dynamic range, and metabolic cost. AI methods can support analysis of high-throughput imaging and response data and can aid computational screening of candidate structures during sensor and receptor design. Combined with structure-guided evaluation, data-driven signal decoding can increase efficiency and support iterative optimization of environmental sensing systems

597 (Murphy et al., 2024; Notin et al., 2024).

598 **Powering synthetic environments and plant factories**

599 Plant factories and vertical farming systems are not breeding platforms in the
600 traditional sense, but their controlled light, temperature, humidity, and nutrient
601 regimes provide a strong foundation for accelerated breeding, trait testing, and
602 iteration. As demand grows for sustainable agriculture and resilient food systems,
603 controlled environment agriculture is becoming an important bridge between trait
604 design, performance evaluation, and optimization. At the same time, enclosed
605 production systems impose new adaptation requirements. Crops must sustain high
606 productivity and efficient resource use while producing stable and predictable
607 phenotypes under defined light spectra, nutrient formulations, and cultivation regimes.
608 Protein engineering can provide a molecular basis for programmable traits tailored to
609 these conditions.

610 Engineering of regulatory proteins and signaling components can equip plants with
611 adaptive responses to facility-specific cues. For example, light- or temperature-
612 responsive transcription factors could be engineered to activate growth or defense
613 pathways under defined conditions. PULSE is an optogenetic system that is inactive
614 under ambient white light but activated by red light. In transgenic *Arabidopsis*,
615 PULSE was used to control immune responses and synthetic signaling, demonstrating
616 application of protein design for tailored sensing and response (Ochoa-Fernandez et
617 al., 2020). Protein engineering can also support modular expression systems that use
618 standardized promoters, signal peptides, and scaffolds, which can improve
619 consistency of trait expression across genotypes and species (Engler et al., 2014;
620 Hahn et al., 2020).

621 In controlled environments, engineered sensors and actuators can be integrated into
622 synthetic circuits to drive condition-specific expression of key traits (Kassaw et al.,
623 2018). Circuits that regulate flowering, nutrient transport, or stress-response proteins

could activate only under particular light spectra or nutrient levels (Khan et al., 2025). Programmable protein switches responsive to hormones or metabolites can provide additional tuning while limiting growth penalties (Zhou et al., 2021).

Beyond single-layer control, multi-tier regulatory systems that integrate internal physiological state with external inputs could coordinate resource allocation, developmental timing, and defense responses. Such designs could support closed-loop optimization in synthetic environments. AI-assisted protein engineering and circuit modeling are being used to improve design and screening of portable regulatory components and modular circuits, with the goal of strengthening self-monitoring and responsive control in plants (Khan and Lister, 2025; Rai et al., 2024; Zhang *et al.*, 2025). Overall, plant factories provide reproducible settings for testing programmable traits and evaluating engineered circuits under defined constraints. A central challenge is to consider modular components together with crop genetic background, cultivation regime, and energy and resource limits, and to measure stability, dynamic range, and metabolic costs. AI can also help calibrate relationships between environmental inputs, engineered signal processing, and trait outputs under controlled conditions, supporting predictable performance across parameter combinations and continued optimization in synthetic agricultural systems.

Challenges of AI-driven protein engineering in plant trait design

AI-driven protein engineering has improved structure prediction and sequence generation, but translation to plant breeding at scale remains limited. Many reported successes still depend on rational engineering and empirical screening and do not yet meet the requirements of a breeding-grade workflow that is predictive, iterative, and reproducible across settings. The gap reflects a set of coupled constraints, including domain shift in plant-relevant data, reliability limits in generative models, genetic background dependence, multiscale mapping from molecular changes to phenotypes, and low validation throughput with limited end-to-end toolchain integration in plant

systems. Together, these factors make it difficult to convert molecular designs into field-level trait gains and prolong design-build-test-learn cycles.

Plant domain shift and model coverage gaps

Most leading structure prediction and generative design models, including AlphaFold2, RoseTTAFold, ProteinMPNN, and RFdiffusion, are trained mainly on general protein sequence and structure datasets. Training distributions are therefore skewed toward soluble enzymes, conserved metabolic proteins, and proteins from a small set of intensively studied organisms (Baek *et al.*, 2021; Jumper *et al.*, 2021; Watson *et al.*, 2022). Plant proteomes, by contrast, include substantial lineage-specific content and distinctive domain architectures, with many plant-specific proteins and structurally complex families such as immune receptors (Panchy *et al.*, 2016; Yang *et al.*, 2009). Experimentally resolved plant structures remain limited. For example, considering only entries with experimental structures in the Protein Data Bank, coverage of the *Arabidopsis thaliana* proteome has been estimated at about 4% (Burley *et al.*, 2025). In addition, deep mutational scanning is not yet broadly established in plant research (Narayanan *et al.*, 2024), in part because links between mutations and agronomic traits are indirect, multiscale, and strongly influenced by environment. These limits reduce opportunities for plant-specific fine-tuning, calibration, and systematic validation. As a result, model generalization, uncertainty behavior, and error modes for key plant targets remain poorly characterized, which complicates candidate prioritization and risk assessment. The problem is especially acute for plant immune proteins such as nucleotide-binding leucine-rich repeat receptors and for cross-kingdom interaction systems involving pathogen effectors and plant receptors, where accurate domain annotation, conformational switching, and activation dynamics are central to function (Jones *et al.*, 2016; Monteiro and Nishimura, 2018). AlphaFold-based approaches are increasingly used in these contexts, but plant-focused benchmarking is still needed to define when predictions are reliable and where systematic biases arise (Lin *et al.*, 2025).

Generative hallucination and biological feasibility gaps

Generative models can propose sequences and folds that score well under internal objectives and appear structurally consistent *in silico*. However, optimizing a model score does not guarantee expression, correct folding, or function in plant cells, which creates a recurring risk of designs that look plausible computationally but fail biologically (Hsu et al., 2024; Listov *et al.*, 2024). In *de novo* design, structures produced by neural-network-based hallucination or diffusion approaches often fail during experimental testing because of folding kinetics, expression burden, or incompatibility with cellular conditions (Anishchenko et al., 2021; Watson et al., 2023). In plants, this gap can be larger because function depends on compartmentalization and local microenvironments and often requires specific assembly factors, post-translational modifications, and tissue- or stage-specific regulation (Jarvis, 2008; Millar et al., 2019). These contextual requirements are not represented explicitly in most current generative models, so designs that look strong by computational criteria can still misfold, mislocalize, or lose function in *planta*.

Several studies therefore emphasize that generative design cannot rely on single-model scores or structural plausibility alone (Chica and Ferruz, 2024; Hsu *et al.*, 2024; Listov *et al.*, 2024). More robust practice requires explicit constraints in the objective, multistage filtering with independent checks, and uncertainty-aware calibration to flag high-risk candidates and reduce false confidence.

Genetic background dependence and limited portability

Plant breeding operates across diverse germplasm. The same engineered protein or edited allele can show strong background dependence and variable effects across cultivars or inbred lines. Differences can arise from variation in regulatory network structure, metabolic context, protein interaction partners, and expression stability (Mackay, 2014). Genetic background can also interact with factors such as genomic integration site to shape transgene expression, so observed phenotypes may not be explained by construct design alone (Alonso-Blanco et al., 2009; Gelvin, 2017). As a result, designs that perform well in one background may not translate into stable gains

across broader breeding materials (Eguchi et al., 2019; Undurraga et al., 2012).

From a modeling perspective, generative approaches trained on within-species or closely related diversity can learn higher-order residue dependencies and epistatic constraints. DeepSequence, for example, captures such dependencies and can be used to assess how mutational effects vary across sequence backgrounds (Riesselman et al., 2018). Related evolutionary generative models, including EVE and EVEscape, represent site dependencies to estimate how effects shift with background (Thadani et al., 2023). These approaches offer a route to quantify background-dependent risk during design and to prioritize variants with greater cross-background robustness. For plant applications, portability and robustness should be treated as explicit engineering objectives and incorporated through constraint-aware design and calibration, rather than deferred to post hoc validation, to support translation and scalable deployment (Chica and Ferruz, 2024; Krapp et al., 2024; Rosace et al., 2023).

The genotype-to-phenotype gap and limited field predictability

AI-driven protein engineering can optimize molecular properties such as stability, binding affinity, substrate specificity, and in vitro catalytic activity (Hsu *et al.*, 2024; Yang *et al.*, 2019). However, agronomic traits result from complex, multiscale genetic and physiological networks, influenced by genotype-by-environment interactions, making them much harder to predict. This creates a substantial genotype-to-phenotype gap between molecular perturbations and field-level performance (Van Eeuwijk et al., 2019; Xu et al., 2022). The gap is particularly evident for protein-level modifications because improvements in a single protein are often buffered by factors such as flux reallocation, network compensation, or physiological regulation (Moreno-Sánchez et al., 2008; Yin and Struik, 2017). For instance, expression of a cyanobacterial Rubisco with higher catalytic turnover in *Tobacco* increased carbon fixation potential *in vitro* but did not lead to improved photosynthetic rate or biomass under ambient carbon dioxide. System-level constraints on CO₂ supply and regulation limited the translation

to whole-plant performance (Lin et al., 2014). This example shows that protein-level optimization alone may not yield agronomic benefits if broader system constraints are not addressed.

To bridge the genotype-to-phenotype gap, it is crucial to integrate protein changes into multiscale models that account for genetic, physiological, and environmental factors (Technow et al., 2015; Wu et al., 2016). Multimodal AI can be a powerful tool in this process by combining protein-level features with transcriptomic, metabolomic, and environmental data to learn nonlinear relationships between molecular changes and phenotypic outcomes (Acosta et al., 2022; Wang et al., 2024; Xu *et al.*, 2022). Additionally, systems biology approaches can help by integrating protein changes, such as modifications to enzyme kinetics, into genome-scale metabolic or process-based physiological models. These models can predict how changes at the molecular level impact broader plant systems, including flux partitioning, resource use, and overall physiological performance. This can facilitate *in silico* prioritization of genetic modifications before costly experimental work (Beilsmith et al., 2022; Yin and Struik, 2017). Finally, iterative design-build-test-learn cycles that incorporate controlled environment or field phenotyping are essential for recalibrating models, ensuring predictive reliability, and improving generalization across different environmental conditions and genetic backgrounds for complex traits.

End-to-end workflow bottlenecks in plant design-build-test-learn

Even when AI models generate strong candidate designs, the build and test stages remain the main constraint on iteration speed and scale in plant systems (Rajput et al., 2025). Stable transformation, greenhouse evaluation, and multi-season, multi-location field trials are slow, costly, and low throughput, which limits feedback and slows iterative improvement (Anjanappa and Gruijssem, 2021; Cobb et al., 2013). Regulatory and biosafety requirements can further extend timelines from design to deployment. In addition, spatiotemporal heterogeneity in field environments increases phenotypic noise and complicates standardization, which weakens feedback for model learning

(Cobb *et al.*, 2013). Software tools and data platforms in plant research also face adoption barriers and limited interoperability. Lack of auditable, reproducible workflows continues to limit scaling.

To address these constraints, we propose a staged roadmap for AI-driven plant trait design. In Stage I, we implement AI-assisted decision tools to support prioritization of candidates and streamline the testing process, reducing the burden of experimental iterations. Additionally, Stage II will integrate multimodal modeling and automated workflows to enhance scalability, ensuring faster iterations and more efficient handling of field and environmental data.

Addressing these constraints requires end-to-end workflow engineering rather than isolated fixes. Intermediate platforms, including protoplast transient expression and *Agrobacterium*-mediated transient assays, can support early screening and risk filtering with rapid readouts of expression, localization, and partial functional proxies (Sparkes *et al.*, 2006; Yoo *et al.*, 2007). These systems do not capture long-term development or environment-dependent regulation and are therefore best used for front-end triage rather than final evaluation (Chupeau *et al.*, 2013; Xu *et al.*, 2021). Data-driven experimental design, including active learning, can prioritize candidates with high expected information gain or trait potential for the next round, increasing learning per assay and accelerating convergence (Yang *et al.*, 2025b). In parallel, standardized, auditable workflows aligned with FAIR principles are needed to reduce tool and data silos and improve reproducibility and portability, which would shorten iteration cycles and support broader deployment. Taken together, AI-driven breeding agent frameworks that coordinate model decisions, experimental validation, and data infrastructure represent a plausible direction for scaling AI-driven protein engineering in plants (Sarkar *et al.*, 2024; Yang *et al.*, 2025a; Zhu *et al.*, 2024).

Perspective: a staged roadmap for plant trait design with AI-driven protein engineering

AI-driven protein engineering is emerging as an enabling technology with the potential to affect multiple areas of plant breeding (Farooq et al., 2024; Zhang *et al.*, 2025; Zhu *et al.*, 2024). Compared with approaches that rely on empirical discovery and labor-intensive screening, protein engineering supports more programmable and scalable molecular modification. Advances in deep learning, structural biology, and computational modeling now allow more precise manipulation of protein function and can support development of designed traits with more predictable and tunable outcomes. These capabilities can shorten breeding cycles and expand the range of candidate traits beyond what is readily accessible from natural variation. In principle, they can support goals such as stress resilience, yield improvement, and higher nutrient use efficiency (Gao, 2021; Rodríguez-Leal et al., 2017; Tian et al., 2021). As AI methods, synthetic biology, and higher-throughput experimental platforms converge, protein engineering could become a major component of crop improvement strategies aimed at sustainable agriculture and food security under climate change and population growth (Eckardt et al., 2023; Tian *et al.*, 2021).

However, translation of molecular designs into breeding-grade outcomes remains limited by several linked bottlenecks, and progress is likely to follow a stepwise trajectory. Based on current technical readiness and validation capacity, we propose a three-stage roadmap (Figure 3). The roadmap describes a progression from AI-supported decision tools to more integrated automation and, eventually, agent-like systems. For each stage, it outlines key tasks, technical challenges, and feasible milestones.

Stage I (near term, 0–3 years): AI-assisted decision support for protein engineering and breeding

In Stage I, workflows remain human-led, with AI used to support decisions within existing protein engineering and breeding pipelines. The goal is to improve design efficiency, reduce trial and error, and increase validation success rates. AI acts mainly as a decision support layer for prioritizing candidates, compressing mutational search

space, and filtering risk, rather than replacing researchers with end-to-end automation (Chica and Ferruz, 2024; Kortemme, 2024; Yang *et al.*, 2019). At this stage, systems operate under close supervision and do not run fully autonomous closed loops. Representative capabilities include prediction of function, stability, and interactions from sequence and structure information (Baek *et al.*, 2021; Homma *et al.*, 2023; Jumper *et al.*, 2021). AI tools can also score and prioritize large mutational libraries or generative candidates (Dauparas *et al.*, 2022; Riesselman *et al.*, 2018). Reliability screening can use multi-model agreement, uncertainty estimation, and confidence calibration to filter higher-risk designs (Edmunds *et al.*, 2024; Kovalevskiy *et al.*, 2024). Constraint-aware approaches can compress mutational space and recommend combinatorial sets of variants to reduce experimental burden while improving hit rates (Ding *et al.*, 2024a; Wu *et al.*, 2019).

Running in parallel, several engineering foundations are immediately actionable. First, benchmark datasets for plant targets and complexes are needed to quantify domain shift, support calibration, and enable standardized evaluation. Second, tiered validation systems, including protoplast assays and transient expression, should be formalized as front-end screening modules with clearly defined limits and explicit handoffs to longer-term evaluation. Third, improved data standards, phenotyping protocols, and traceable workflows are needed to strengthen reproducibility and cross-team portability and to prepare for faster design-build-test-learn cycles in later stages.

Stage II (mid term, 3–7 years): AI-integrated multi-module systems with limited autonomy

Stage II shifts AI from decision support to an integrated system that coordinates multiple model-driven decisions across design-build-test-learn cycles with limited autonomy. The defining feature is integration across modules under explicit constraints and governance guardrails, enabling semi-autonomous closed-loop iteration within predefined boundaries. Humans still specify goals, approve plans, oversee execution, and make final decisions (Häse *et al.*, 2019; Martin *et al.*, 2023). This stage targets the

experimental feedback bottleneck, scaling limits, and genotype-to-phenotype mapping challenges described above.

Key elements include a multimodal modeling framework that combines protein structural features with genomic, transcriptomic, metabolomic, and phenomic data to build multiscale predictors that link molecular variation to field-relevant traits and shorten the genotype-to-phenotype inference chain (Van Eeuwijk *et al.*, 2019; Xu *et al.*, 2022). Active learning and Bayesian optimization can then prioritize experiments and guide exploration of design space under resource constraints, accelerating discovery and validation of functional variants (Yang *et al.*, 2025b). In addition, integration with automation infrastructure, including high-throughput phenotyping, biofoundries, and automated molecular experimentation, can support scalable, reproducible wet and dry closed-loop operation (Fahlgren *et al.*, 2015; Hillson *et al.*, 2019).

Within this architecture, the system can support complex breeding goals, such as jointly improving yield and stress resilience, by proposing multiplex genome editing strategies, forecasting expression stability across environments, and coordinating experimental resources for rapid iteration (Cheng *et al.*, 2025; South *et al.*, 2019). For example, to improve photosynthetic efficiency, it could design multiple enzyme variants in parallel and use automated platforms to build and test them concurrently, then recommend the best-performing combination (Kromdijk *et al.*, 2016). Reaching this stage will require progress in causal modeling for integrative plant multi-omics, standardized interfaces to high-throughput functional validation platforms, and auditable human AI decision interfaces that support oversight and compliance.

Stage III (long term, 7–10 years): Development of AI-driven bio-breeding agents – A conceptual framework

Generalist protein engineering agents such as EvoPlay, SPARKS, and Virtual Lab have shown that fully automated, end-to-end design is feasible in several biological settings (Ghafarollahi and Buehler, 2025; Ray *et al.*, 2012; Wang *et al.*, 2023). These

platforms can generate functional proteins with high precision and limited human input. Application in plants remains untested, and while current methods in enzyme engineering show potential, significant challenges remain before they can be adapted for plant breeding. For example, an autonomous enzyme engineering platform improved activity of *Arabidopsis thaliana* halide methyltransferase through iterative design, testing, and optimization using automated cycles of variant generation and screening in a biofoundry environment (Singh et al., 2025). This example indicates that plant enzymes can be tractable targets for similar workflows. While these systems show potential for routine, on-demand protein innovation in non-plant systems. However, their application to plant breeding remains a long-term objective requiring substantial technological advancements.

A potential next step is the development of AI-driven breeding agents, which may eventually couple protein engineering with genome editing and multi-omics analysis (Sarkar et al., 2024; Yang et al., 2025a; Zhu et al., 2024). Full integration and autonomy remain long-term objectives, requiring significant technological advancements. In this conceptual framework, an AI-driven agent would be integrated with biotechnology toolsets to provide guidance from the gene and protein level through to breeding applications, although full integration and autonomy remain distant goals. The goal is tighter alignment between molecular design and practical crop improvement.

One potential future direction is the development of an AI-driven bio-breeding agent, conceptualized as six interconnected modules (**Supplemental Figure 2**). First, a multimodal model trained on species-specific data provides a shared representation that integrates genomic information, protein structure, multi-omics state, and phenotypic readouts, supporting inference from molecular perturbations to trait outcomes. Second, a knowledge graph and gene-to-phenotype association network integrates relationships among genes, proteins, traits, and environments, supporting target decomposition, candidate prioritization, and editing strategy design. Third, a structure and function module combines structure prediction, interaction modeling, and mutational effect

assessment to identify molecular determinants and constraints. Fourth, an AI plus biotechnology design and optimization module coordinates generative models with engineering interventions, including rational and semi-rational redesign and *de novo* design, to generate trait-oriented molecular solutions. Fifth, a breeding decision module connects molecular outputs with genomic selection, genotype-to-phenotype modeling, and genotype-by-environment analysis to support optimization across traits and environments. Sixth, a human AI interaction and workflow orchestration module provides interfaces, monitoring, and auditable workflows so humans set objectives, manage risk, and interpret results.

AI-driven protein engineering and future breeding paradigms

AI-driven protein engineering is changing the methodological basis of plant breeding. The staged roadmap above suggests a stepwise trajectory that remains human centered, collaborative, and governed by safeguards from the outset. In the near term, priorities include AI-supported decision tools and stronger data and benchmarking infrastructure. In the mid term, emphasis shifts to semi-autonomous integrated systems capable of sustained, constrained iteration. In the longer term, progress will depend on scientific prerequisites and governance conditions that allow responsible development of more autonomous agents. If achieved, this progression could shorten breeding cycles and expand the space of designable traits. It could also drive a broader shift in breeding practice, from empiricism toward model-driven reasoning, from single-trait improvement toward systems-level design, and from local optimization toward outcomes evaluated against sustainability constraints. As climate change and food security pressures intensify, AI-driven protein engineering could become a major contributor to intelligent breeding and long-term agricultural sustainability.

Competing interests

The authors declare no competing interests.

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Author contributions

X.W. and R.F. conceived the project. R.F. and S.J. drafted the manuscript. C.Y., T.W., X.W., and J.Y. reviewed and revised the manuscript. All authors reviewed and approved the final version.

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Figure 1 | Evolution of protein engineering and protein design methodologies.

Conceptual overview of the development of protein engineering, with protein design as a central methodological stream. The diagram shows progression from early rational design based on experimentally determined structures and mechanistic knowledge, through semi-rational strategies that combine rational guidance with mutational scanning and directed evolution, to modern AI-driven and *de novo* design approaches. Progress along this continuum has been driven by advances in computational tools and expansion of sequence, structure, and functional datasets, improving efficiency, scalability, and predictive performance. The figure also summarizes representative workflows. In rational and semi-rational design, target structure determination and mutation design comprise the design stage, while protein expression and screening or selection are typically combined as build and test. In *de novo* design, the workflow is shown in a design-build-test-learn framework, where target structure specification, backbone generation, and sequence design define design; protein expression corresponds to build; validation, including data-based and experimental assessment, defines test; and model updating based on validation outcomes defines learn, enabling iterative optimization.

Figure 2 | Application scenarios of protein engineering in plant breeding.

Overview of eight representative scenarios where protein engineering supports trait improvement, with AI contributing at different stages of analysis and optimization. These include: A. Enhanced disease resistance through engineered immune receptors; B. Improved insect resistance through optimization of insecticidal proteins; C. Increased abiotic stress tolerance through stabilization and functional tuning of metabolic enzymes; D. Improved nutrient uptake and resource-use efficiency through redesign of transporters; E. Variant mining and trait fine-tuning supported by predictive modeling; F. Optimization of genome editing systems using engineered nucleases and precision editors; G. Environmental sensing through synthetic biosensors responsive to endogenous signals; and H. Programmable regulatory circuits for synthetic

environments and plant factories. NLR, nucleotide-binding leucine-rich repeat; ABA, abscisic acid; WT, wild type; CBE, cytosine base editor; ABE, adenine base editor; IAA, indole-3-acetic acid.

Figure 3 | A staged roadmap for plant trait design with AI-driven protein engineering.

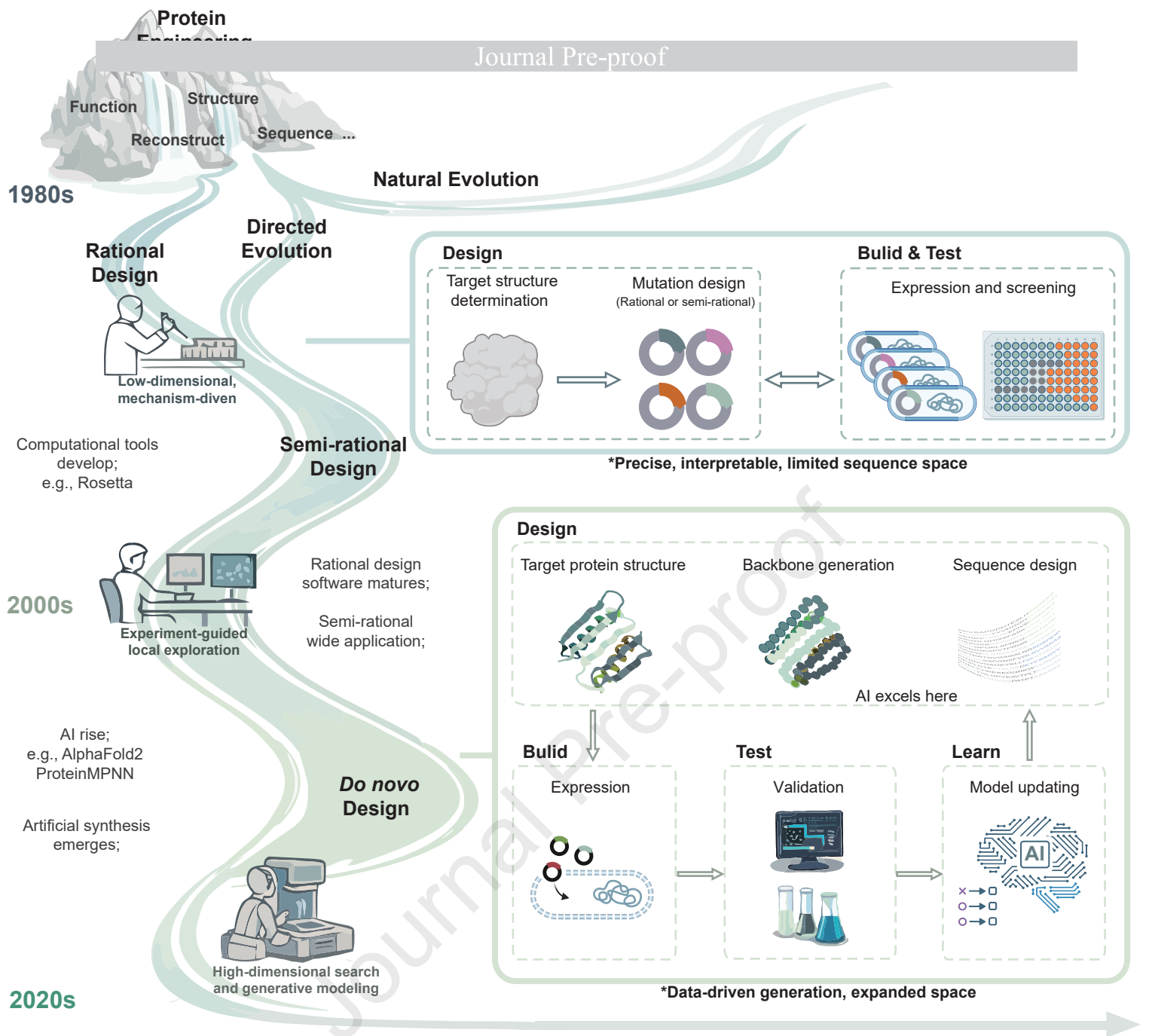
Three-stage trajectory for deploying AI-driven protein engineering in plant breeding, organized by time horizon, system autonomy, and functional scope. Stage I (0–3 years) focuses on human-led, AI-assisted decision support, where AI tools are gradually integrated into existing workflows to improve predictions of molecular function, stability, and interactions, prioritize candidate variants, and filter higher-risk designs. Key deliverables include plant-specific benchmarks, tiered prescreening validation stacks, and FAIR, traceable workflows. Stage II (3–7 years) describes multi-module integrated systems with limited autonomy under explicit guardrails, where AI coordinates model-driven decisions across design-build-test-learn (DBTL) cycles. This stage integrates multimodal genotype-to-phenotype (G2P) modeling, active learning, and automated experimentation, with milestones including reproducible DBTL loops, experiment scheduling, and auditable automation. Stage III (7–10 years) is conceptual and governance-first, describing a long-term vision for AI bio-breeding agents capable of end-to-end workflow orchestration. These systems aim to link molecular design to field-level outcomes through six core functional modules, contingent on unresolved scientific prerequisites and governance frameworks that ensure reliability, transparency, and compliance. AL/BO, active learning and Bayesian optimization; FAIR, findable, accessible, interoperable, and reusable.

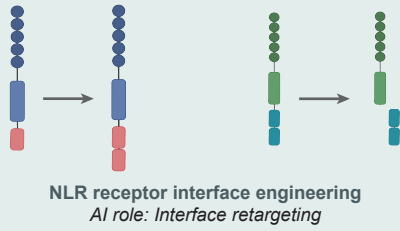
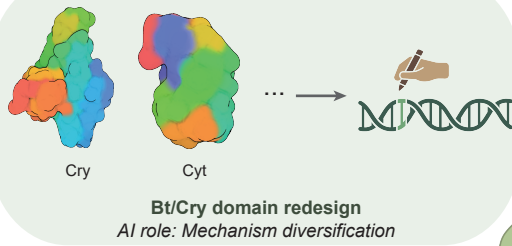
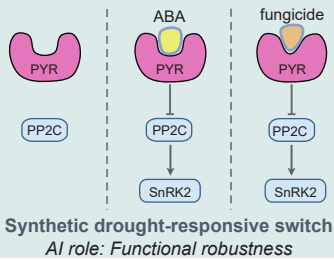
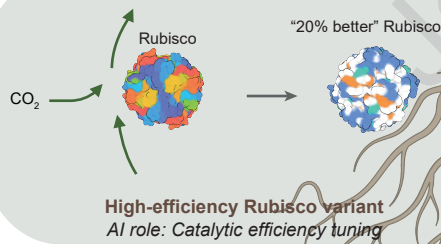
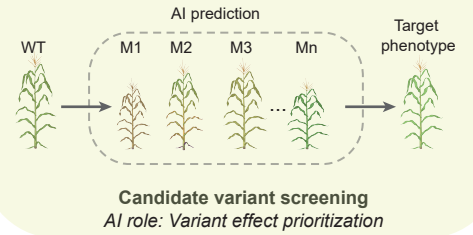
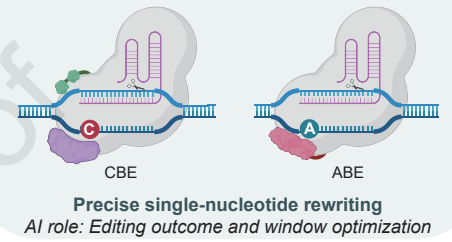
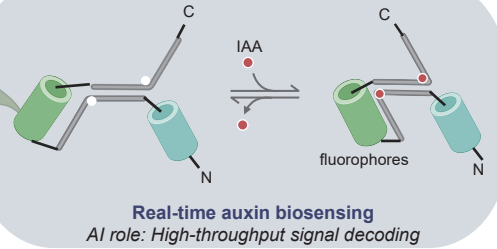
Supplemental Figure 1 | Future applications of *de novo* design in plant breeding.

Schematic showing how inputs such as target fold, functional site, desired activity, or binding partner can guide generation of proteins with tailored molecular properties for crop improvement.

Supplemental Figure 2 | Schematic of infrastructure for an AI-driven bio-breeding agent platform.

Schematic of a proposed AI-driven bio-breeding agent platform that integrates AI with biotechnology for crop improvement. A human-machine interaction and task monitoring system sits at the center, providing an interface for submitting breeding tasks, setting parameters, and monitoring analysis. Surrounding modules are connected in a logical workflow. Species-specific multimodal models integrate genomic, proteomic, and metabolomic data through pretraining and fine-tuning, providing shared representations for downstream modules. A species-specific gene-to-phenotype knowledge graph stores structured relationships among genes, phenotypes, pathways, and curated resources to support planning, protein interpretation, and design. A protein structure and function module supports structure parsing, prediction of interaction hotspots, identification of catalytic sites, and assessment of mutational effects. An AI plus biotechnology (AI+BT) protein design module supports rational and semi-rational redesign, refactoring, and *de novo* design. A breeding application module integrates genomic selection (GS), genotype-to-phenotype (G2P) analysis, and gene-by-environment ($G \times E$) modeling to support trait prediction and breeding optimization. Together, these components aim to connect molecular design to breeding decisions and accelerate trait development while accounting for stress tolerance and nutrient efficiency.



A Enhances disease resistance**B Reinforces insect resistance****C Boosts abiotic stress resilience****D Improves efficiency and nutrition****E Variant mining and trait fine-tuning****F Optimizes genome editing components****G Programs environmental sensing systems****H Powers synthetic environments and plant factories**